

# Graph Inequalities

Lesson 7-5

**Name:** \_\_\_\_\_ **Date:** \_\_\_\_\_ **Class:** \_\_\_\_\_

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## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can graph the solutions of an inequality on a number line.

**Language Objective:** I can explain my graph using the words number line, open circle, closed circle, and solution set.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Number line

Open circle

Closed circle

Solution set

Inequality

Boundary point



Tap any word to see what it means and a picture.

1

A line with numbers in order used to show values and solutions is a

2

A hollow circle that shows a value is NOT included, used for  $<$  or  $>$ , is an

3

A filled-in circle that shows a value IS included, used for  $\leq$  or  $\geq$ , is a

4

All the values that make an inequality true form the

5

A math sentence comparing two amounts with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$  is an

- 6 The number where the solution set begins on the number line is the

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**When you graphed  $x > 4$  and  $x \leq 7$ , why did you use an open circle on one and a closed circle on the other?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.



**Number line** — A line with numbers in order. You use it to show answers.

*Recta numérica — Una línea con números en orden. La usas para mostrar respuestas.*



**Open circle** — An empty circle showing a number is NOT included.

*Círculo abierto — Un círculo vacío que muestra que un número NO está incluido.*



**Closed circle** — A filled-in circle showing a number IS included.

*Círculo cerrado — Un círculo relleno que muestra que un número SÍ está incluido.*



**Solution set** — All the numbers that make the inequality true.

*Conjunto solución — Todos los números que hacen verdadera la desigualdad.*



**Inequality** — A math sentence comparing two amounts with  $<$ ,  $>$ ,  $\leq$ , or  $\geq$ .

*Desigualdad — Una oración matemática que compara dos cantidades con  $<$ ,  $>$ ,  $\leq$  o  $\geq$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**I used a \_\_\_\_ circle because the symbol \_\_\_\_ means the boundary is \_\_\_\_.**

*Usé un círculo \_\_\_\_ porque el símbolo \_\_\_\_ significa que el límite \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Because 2:00 itself counts, I use a closed circle at 2.

**Evidence:** Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy  $t \geq 2$ .

**Reasoning:** This evidence connects the definition of number line to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- I used a \_\_\_\_ circle because the symbol \_\_\_\_ means the boundary is \_\_\_\_.

*Usé un círculo \_\_\_\_ porque el símbolo \_\_\_\_ significa que el límite \_\_\_\_.*

- I shaded \_\_\_\_ because the solution set includes \_\_\_\_.

*Sombree \_\_\_\_ porque el conjunto solución incluye \_\_\_\_.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

- I know because \_\_\_\_.

*Lo sé porque \_\_\_\_.*

- So \_\_\_\_.

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*